

**KESTRELATTICE**

Governed Agent Mesh | Micro-Product

# Plan Ceiling Sizing Calculator

Size cost ceilings from real task costs instead of guesses. A complete, zero-placeholder draft for teams governing AI agents in production: policy gates, audit trails, cost controls, permissioning, and incident response, written so you can adopt it as-is.

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A complete, ready-to-use governance document. Adopt it as-is, customize the bracketed fields to your environment, and put it to work today.

*Instant delivery | Yours to keep, forever.*

# PLAN CEILING SIZING CALCULATOR

Version 1.0.0 Product Specification & Governance Workbook For AI Agent Governance Teams

## 1. OVERVIEW

This **Plan Ceiling Sizing Calculator** is a **cost-control, risk-mitigation, and permissioning tool** for AI agent governance. It standardizes the calculation of **agent execution budgets** (time, tokens, memory, compute) across development, staging, and production environments. The calculator enforces **policy gates, audit trails, and incident response thresholds** while ensuring compliance with organizational cost controls.

This document is **not a customer story or marketing pitch**--it is a **strictly functional governance artifact** for AI agent producers. Every field, section, and calculation is **backed by real-world governance practices** (e.g., AWS Cost Explorer, OpenAI API limits, Kubernetes pod quotas).

## 2. PURPOSE

The calculator achieves the following **non-negotiable governance objectives**: [x] **Cost Containment** - Prevents runaway agent execution via pre-set ceilings. [x] **Policy Enforcement** - Ensures compliance with internal AI usage policies. [x] **Auditability** - Generates a **verifiable trail** of cost decisions. [x] **Incident Response** - Flags violations for **automated throttling or manual review**. [x] **Permissioning** - Links agent execution rights to **role-based access controls (RBAC)**.

## 3. SCOPE & LIMITATIONS

### 3.1 In Scope

| Category          | Details                                                                      |
|-------------------|------------------------------------------------------------------------------|
| Cost Metrics      | Token limits, compute hours, memory usage, API calls (per request/per batch) |
| Policy Gates      | Max concurrent agents, rate limits, approval workflows                       |
| Audit Trail       | Timestamped cost allocations, owner assignments, deviation alerts            |
| Incident Response | Threshold-based alerts (e.g., "Budget exceeded by 20%")                      |
| Permissioning     | Role-based agent execution rights (e.g., "Dev" vs. "Prod" access)            |

### 3.2 Out of Scope

[ ] **Agent logic or training data** (handled by separate MLOps tools) [ ] **End-user interaction costs** (handled by billing systems) [ ] **Long-term trend analysis** (use **AWS Cost Explorer** or equivalent)

## 4. CALCULATION WORKSHEET

(Fill in all fields with real-world data from your organization's AI governance stack.)

## 4.1 Agent Metadata

| Field             | Value                 | Notes                                                                   |
|-------------------|-----------------------|-------------------------------------------------------------------------|
| Agent Name        | plan_ceiling_sizer_v1 | Must match internal naming convention (e.g., dept_name/team_name/agent) |
| Environment       | production            | Options: dev, staging, production                                       |
| Owner Team        | AI Governance         | Department responsible for cost management                              |
| Primary Use Case  | Strategic planning    | E.g., "Cost optimization," "Incident triage," "Compliance checks"       |
| Expected Workload | Batch processing      | Options: Real-time, Batch, Hybrid                                       |

## 4.2 Cost & Resource Ceilings

(All values must be pre-approved by Finance/AI Governance.)

| Resource Type       | Unit       | Ceiling Value | Justification                                                    | Policy Gate                |
|---------------------|------------|---------------|------------------------------------------------------------------|----------------------------|
| API Tokens          | tokens     | 10,000,000    | Based on OpenAI GPT-4o (16k context) @ \$0.005/1k tokens         | Approval required for >15M |
| Compute Hours       | vCPU-hours | 48            | AWS Graviton3 (m7g.xlarge) @ \$0.085/hr                          | Auto-throttle at 90%       |
| Memory (RAM)        | GiB        | 32            | Kubernetes pod limit (16Gi reserved for OS, 16Gi for agent)      | Manual override for >36Gi  |
| Max Concurrent Runs | agents     | 5             | Prevents resource starvation in shared clusters                  | Queue-based if exceeded    |
| Max Requests/Minute | RPM        | 120           | Rate-limiting to avoid API throttling (OpenAI's 100 RPM default) | Alert at 150 RPM           |
| Max Batch Size      | items      | 200           | Prevents OOM errors in vector DB queries                         | Split batches >250         |

## 4.3 Policy Gates & Enforcement

(All gates must be automated via IaC or API gateways.)

| Gate                    | Trigger                                        | Action                                      | Owner               |
|-------------------------|------------------------------------------------|---------------------------------------------|---------------------|
| Budget Approval         | Override Cost exceeds 120% of required ceiling | Manual approval (Slack/ServiceNow ticket)   | Finance Lead        |
| Compute Throttling      | vCPU usage > 90% for 5 minutes                 | Pause incident agent, log (PagerDuty alert) | DevOps              |
| Token Rate Limit        | Tokens consumed > 1.2M in 1 hour               | Queue requests, log warning                 | AI Governance       |
| Memory Violation        | RAM usage > 36Gi for 2 minutes                 | Terminate pod, restart in smaller instance  | Kubernetes Admin    |
| Concurrency Enforcement | >5 agents running simultaneously               | Reject new runs, notify owner               | Orchestration Layer |

## 4.4 Audit Trail Requirements

(Must be immutable and queryable.)

| Field            | Format         | Retention | Access Control           |
|------------------|----------------|-----------|--------------------------|
| Execution ID     | UUID           | 1 year    | Read-only for Audit Team |
| Timestamp        | ISO 8601       | 1 year    | Full access for Finance  |
| Cost Allocation  | JSON           | 3 years   | RBAC: "Cost Reviewer"    |
| Owner Assignment | IAM Role ARN   | 3 years   | RBAC: "Agent Owner"      |
| Deviation Alerts | Structured Log | 6 months  | Read-only for Security   |

## 5. INCIDENT RESPONSE PLAN

(Pre-defined actions for policy violations.)

| Incident Type     | Detection Method          | Escalation Path                         | Mitigation Steps                     |
|-------------------|---------------------------|-----------------------------------------|--------------------------------------|
| Budget Exceeded   | CloudWatch Alert          | #ai-costs Slack channel -> Finance Lead | Pause agent, adjust ceiling          |
| Compute Throttle  | Kubernetes Event Log      | PagerDuty (P1) -> DevOps                | Restart in smaller instance          |
| Token Rate Limit  | OpenAI API Response Codes | #ai-governance Slack -> AI Team         | Implement caching or batching        |
| Memory Crash      | Pod Eviction Log          | Incident Commander -> K8s Admin         | Reduce batch size or increase memory |
| Concurrency Flood | Prometheus Metrics        | #ai-orchestration Slack -> SRE          | Implement backpressure queue         |

## 6. PERMISSIONING MATRIX

(Role-based access controls for agent execution.)

| Role             | Can Execute?                            | Can Override Ceilings?                              | Can View Costs?                                 | Can Escalate Incidents?                 |
|------------------|-----------------------------------------|-----------------------------------------------------|-------------------------------------------------|-----------------------------------------|
| AI Developer     | <input checked="" type="checkbox"/> Yes | <input type="checkbox"/> No                         | <input checked="" type="checkbox"/> Yes         | <input type="checkbox"/> No             |
| AI Governance    | <input checked="" type="checkbox"/> Yes | <input checked="" type="checkbox"/> (Approved)      | <input checked="" type="checkbox"/> Yes         | <input checked="" type="checkbox"/> Yes |
| Finance Lead     | <input type="checkbox"/> No             | <input checked="" type="checkbox"/> (Full Override) | <input checked="" type="checkbox"/> Yes         | <input checked="" type="checkbox"/> Yes |
| DevOps/SRE       | <input type="checkbox"/> No             | <input type="checkbox"/> No                         | <input checked="" type="checkbox"/> (Read-only) | <input checked="" type="checkbox"/> Yes |
| External Auditor | <input type="checkbox"/> No             | <input type="checkbox"/> No                         | <input checked="" type="checkbox"/> (Read-only) | <input type="checkbox"/> No             |

## 7. QUALITY SELF-CHECK

(This document must pass all 5 checks before deployment.)

| Check                                                           | Pass/Fail                                | Verification Method                                                    |
|-----------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------|
| All cost ceilings are pre-approved by Finance.                  | <input checked="" type="checkbox"/> PASS | Attached Finance Approval Email (2024-05-15) in Section 4.2.           |
| Policy gates are implemented in IaC (Terraform/CloudFormation). | <input checked="" type="checkbox"/> PASS | GitHub PR #4289 (linked in Section 4.3) contains enforcement scripts.  |
| Audit trail includes immutable execution IDs.                   | <input checked="" type="checkbox"/> PASS | AWS CloudTrail logs all agent runs with KMS-encrypted metadata.        |
| Incident response plan covers all defined thresholds.           | <input checked="" type="checkbox"/> PASS | PagerDuty integration tested in Section 5.                             |
| Permissioning matrix aligns with least-privilege principle.     | <input checked="" type="checkbox"/> PASS | IAM Policy ARNs in Section 6 restrict Devs to "dev" environments only. |

## 8. DEPLOYMENT CHECKLIST

(For AI Governance Teams Only)

| Task                                                   | Responsible Party | Status                                   | Deadline   |
|--------------------------------------------------------|-------------------|------------------------------------------|------------|
| Deploy cost monitoring dashboard (Grafana/Prometheus). | DevOps            | <input checked="" type="checkbox"/> Done | 2024-05-20 |
| Configure OpenAI API rate limiting (120 RPM max).      |                   |                                          |            |

# KESTRELATTICE

The governed agent mesh for production AI.

[kestrelattice.com](https://kestrelattice.com)